

IoT Light Control Via Mobile App

INTRODUCTION

The IoT Light Control via Mobile App project is a smart home automation system that allows users to control electrical devices remotely using a smartphone. In this project, an ESP8266 NodeMCU microcontroller is connected with the Blynk IoT platform to switch an LED or relay ON and OFF through the internet. This system demonstrates the concept of Internet of Things (IoT), where devices communicate wirelessly for remote monitoring and control.

OBJECTIVE

- To control lights or appliances remotely using a mobile application.
- To understand the working of IoT-based automation systems.
- To interface ESP8266 NodeMCU with the Blynk IoT platform.
- To demonstrate wireless communication using Wi-Fi technology.
- To develop a simple and cost-effective smart home application.

COMPONENTS REQUIREMENTS

ESP8266 NodeMCU

5V Relay Module

Breadboard

Jumper Wires

LED with 220Ω Resistor

USB Cable

Wi-Fi Connection

Android Mobile Phone

WORKING PRINCIPLE

The ESP8266 NodeMCU connects to a Wi-Fi network and communicates with the Blynk cloud server. The user operates a button widget in the Blynk mobile application. When the button is pressed, a command is sent through the internet to the ESP8266 board. The microcontroller processes the command and controls the connected LED or relay module. If a relay is connected, it can switch real electrical appliances ON or OFF remotely. This project enables wireless and real-time control of devices from anywhere.

SOFTWARE USED

- Arduino IDE
- Blynk IoT App
- ESP8266 Board Package

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APPLICATIONS

- Smart home automation
- Remote light control systems
- Wireless appliance management
- IoT learning and educational projects

CODE

```
#define BLYNK_TEMPLATE_ID "TMPL3wY5nlqyp"
#define BLYNK_TEMPLATE_NAME "light control led"
#define BLYNK_AUTH_TOKEN "1xRPDtsL6KS_83ZNy3uPVVXMUQ61cvyD"
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
char ssid[] = "V2022";
char pass[] = "Sheetal2004";
#define RELAY_PIN D1
BLYNK_WRITE(V0) { // Virtual Pin 0 from Blynk app button
  int value = param.asInt();
  digitalWrite(RELAY_PIN, value);
  Serial.println(value ? "Relay ON" : "Relay OFF");
}
void setup() {
  Serial.begin(115200);
  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, LOW);
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
}
void loop() {
  Blynk.run();
}
```